

SnapSentry AI

Autonomous On-Device Multimodal Defense System

Real-Time Deepfake Voice Interceptor • Screen Phishing Guard • Social Engineering Shield

for Snapdragon HP PCs — 100% local, private, and always-on.

< 30 ms

Response Time

1.75 W

Power Draw

45 TOPS

Hexagon NPU

100%

On-Device / Air-Gapped



THE PROBLEM

Cyber Threats Have Gone Real-Time

Generative AI has turned attacks from slow, static scripts into dynamic, real-time assaults — tricking professionals out of credentials in seconds.



Voice Cloning Scams

Audio deepfakes impersonate CEOs, family members, and IT staff — trusted voices become attack vectors.



On-Screen Phishing

Deceptive overlays, malicious QR codes, and fake credential prompts sit right on the user's desktop.



Social Engineering

Live calls pressure victims into sharing 2FA tokens or installing remote-desktop tools mid-conversation.

⚠️ Why this matters:

Detection is not enough — the window to act is under 50 ms. Every second of cloud latency is a second the scammer owns.

Why Cloud Security Fails Here



The Privacy Barrier

Continuous desktop video and conference audio can never be streamed to third-party cloud APIs — corporate policy forbids it. Protection must live on-device.



The Latency Penalty

Voice-cloned extortion and urgent 2FA theft demand detection in < 50 ms. Cloud round-trips add 500–2000 ms — far too late to intervene.



The Thermal & Battery Wall

Continuous vision + audio on x86 CPUs burns 25–45 W, spins up fans, and kills battery in under 90 minutes. Only a dedicated NPU sustains this workload all day.

THE VERDICT

Cloud can't protect the live microphone and screen.

CPU can't sustain it.

Only the Snapdragon Hexagon NPU — 45 TOPS at laptop power — makes real-time, always-on, private endpoint defense possible.

THE SOLUTION

SnapSentry AI: An Unbreachable, Air-Gapped Shield

One tri-model pipeline, orchestrated entirely on the Hexagon NPU — continuous protection with zero data ever leaving the device.

VOICERADAR

Deepfake Voice Interceptor

Conformer / Whisper-Tiny (W8A8) analyzes incoming audio for vocoder artifacts and synthetic harmonic phase anomalies — sub-20 ms per buffer.

VISUAL INTERCEPTOR

Screen Phishing Guard

YOLOv11-Nano / MobileNet (INT8) scans active desktop frames at 15 FPS, flagging phishing overlays, malicious QR codes, and command-prompt injections.

CONVERSATIONAL GUARD

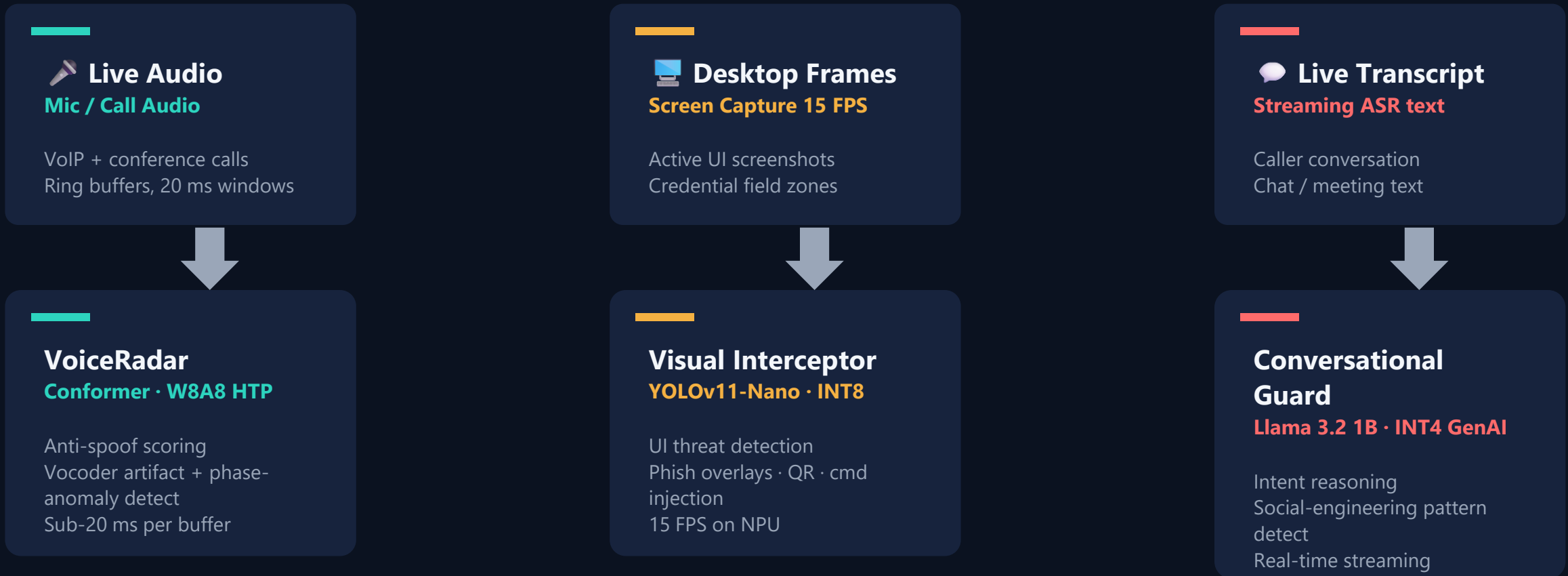
Social Engineering Shield

Llama 3.2 1B / Phi-3.5-mini (INT4, QNN GenAI) parses live transcripts to detect coercion patterns — 2FA demands, remote-desk installation requests.

EXECUTION STACK

Qualcomm AI Hub models → ONNX Runtime with QNN Execution Provider → Hexagon HTP backend → Snapdragon X Elite / Plus (HP OmniBook)

Tri-Model Pipeline on the Hexagon NPU



ZERO-TRUST ACTION LAYER

🚨 Voice-clone warning • 🟥 Deceptive-field bounding & highlight • 🛡️ OTP field blur + clipboard masking

Model Stack — Quantized for the NPU

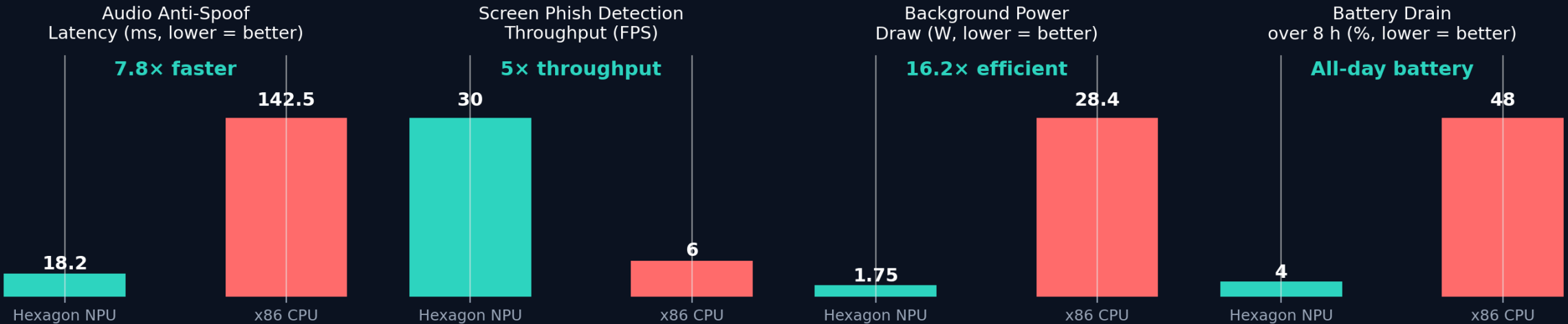
Component	AI Hub Model	Quantization	NPU Role / Functionality
VoiceRadar	Conformer / Whisper-Tiny	W8A8 (HTP)	Anti-spoofing — vocoder artifacts & synthetic harmonic phase anomalies, sub-20 ms per audio buffer.
Visual Interceptor	YOLOv11-Nano / MobileNet	INT8	Active UI scanning at 15 FPS — flags phishing overlays, malicious QR codes, command-prompt injections.
Conversational Guard	Llama 3.2 1B / Phi-3.5-mini	INT4 · QNN GenAI	Real-time transcript intent reasoning — detects 2FA-token demands, remote-desk installs, extortion patterns.

Design principle:

every model is pulled from the Qualcomm AI Hub, pre-quantized (INT4–W8A8), and compiled once for the Hexagon HTP backend — no retraining, no cloud dependency.

PERFORMANCE BENCHMARKS

Hexagon NPU vs. Traditional x86 (AVX2)



The moat:

sub-30 ms end-to-end response at 1.75 W — protection that is imperceptible to the user, silent on fan acoustics, and invisible on the battery meter. An x86 CPU burns 16x the power for 6x less throughput.

Silent Sentinel → Deterministic Zero-Trust Mitigation

ALWAYS-ON:

Runs silently in the Windows system tray at negligible power (< 1.8 W) — until a threat is detected.

1

Synthetic Audio Warning

If an incoming caller's voice exhibits AI-cloning artifacts (vocoder signatures, phase anomalies), a persistent on-screen alert warns the user immediately — before any credentials are shared.

2

Deceptive Phish Bounding

Deceptive password inputs, credential overlays, and malicious QR codes are highlighted in high-contrast red bounding boxes — visually intercepting the click before it happens.

3

Intent Intercept & Safeguard

When the local Llama 3.2 engine flags extortion intent (2FA demands, terminal script execution), it instantly blurs OTP credential fields and masks the Windows clipboard.

Why SnapSentry Wins on Snapdragon

Unbreachable Privacy

Zero data egress — audio, screen, and transcripts never leave the laptop. Air-gapped by architecture, not by policy.

Real-Time by Physics

Sub-30 ms on-device inference beats any cloud round-trip (500–2000 ms) by an order of magnitude.

All-Day Protection

1.75 W background draw = < 4% battery over 8 hours. No fans, no throttle, no user friction.

Built on Qualcomm AI Hub

Production-ready Conformer, YOLOv11-Nano, and Llama 3.2 1B models, pre-optimized for Hexagon HTP via ONNX Runtime QNN EP.

THE VISION

SnapSentry AI turns every Snapdragon HP PC into a self-defending endpoint — making deepfake voice fraud, on-screen phishing, and social engineering economically worthless for attackers.